

Mother's Day Crossword Solutions

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1 8	2 1
3 4	9

ONE ACROSS

WLOG, let $a \leq b \leq c$. Then $a \leq \frac{19}{3}$, so $a \leq 6$. For a fixed value of a , we have

$$b + c = 19 - a$$

$$bc = 99 - a(b + c) = 99 - a(19 - a) = a^2 - 19a + 99.$$

Thus b, c are roots of

$$x^2 - (19 - a)x + (a^2 - 19a + 99) = 0.$$

The discriminant must be a perfect square, meaning that

$$\Delta = (19 - a)^2 - 4(a^2 - 19a + 99) = -3a^2 + 38a - 35$$

must be a perfect square. Testing $a = 1, 2, 3, 4, 5, 6$, we get $\Delta = 0, 29, 52, 69, 80, 85$. The only perfect square is 0, which occurs when $a = 1$. Therefore, $b = c = \frac{19-1}{2} = 9$. Hence, $abc = 1 \cdot 9 \cdot 9 = \boxed{81}$.

ONE DOWN

We need the last two digits, so we work modulo 100. First, $12^{2026} \equiv 0 \pmod{4}$, since 12^n is divisible by 4 for $n \geq 1$. Next, work modulo 25. Since $\varphi(25) = 20$ and $2026 \equiv 6 \pmod{20}$, we have $12^{2026} \equiv 12^6 \pmod{25}$. Now $12^2 \equiv 144 \equiv 19 \pmod{25}$, so $12^4 \equiv 19^2 \equiv 361 \equiv 11 \pmod{25}$. Therefore, $12^6 \equiv 12^4 \cdot 12^2 \equiv 11 \cdot 19 \equiv 209 \equiv 9 \pmod{25}$. So the answer x satisfies

$$x \equiv 0 \pmod{4}$$

$$x \equiv 9 \pmod{25}.$$

Write $x = 9 + 25k$. Since $x \equiv 0 \pmod{4}$, $9 + 25k \equiv 0 \pmod{4}$. This gives $1 + k \equiv 0 \pmod{4}$, so $k \equiv 3 \pmod{4}$. Taking $k = 3$, $x = 9 + 75 = \boxed{84}$.

TWO DOWN

Suppose the tiling uses k heart tiles. Since each heart tile has length 2, the number of rose tiles is $8 - 2k$. To prevent two heart tiles from touching, we place the heart tiles into the gaps around the rose tiles. If there are $8 - 2k$ rose tiles, then there are $8 - 2k + 1 = 9 - 2k$ gaps where heart tiles can go. Since no two heart tiles may touch, we choose k of these gaps:

$$\binom{9 - 2k}{k}.$$

We sum over all possible k to get the answer:

$$\binom{9}{0} + \binom{7}{1} + \binom{5}{2} + \binom{3}{3} = 1 + 7 + 10 + 1 = \boxed{19}.$$

THREE ACROSS

Place the isosceles trapezoid on the coordinate plane with lower base 10 and upper base 4, so that its four vertices are

$$(-5, 0); (5, 0); (-2, h); (2, h).$$

One diagonal has slope $\frac{h}{7}$, while the other has slope $-\frac{h}{7}$. Since the diagonals are perpendicular, the product of their slopes is -1 :

$$\frac{h}{7} \cdot \left(-\frac{h}{7}\right) = -1.$$

Thus, $h^2 = 49 \Rightarrow h = 7$. The area is then

$$\frac{10 + 4}{2} \cdot 7 = 7 \cdot 7 = \boxed{49}.$$